

5. Dynamics

Monday, 10 May 2021 10:48 PM

1 Newton's laws and relativity

- 1) Newton's first law was built into principles of relativity
- 2) Newton's 2nd Law remains valid as long as we use relativistic momentum

$$\left\{ \vec{F} = \frac{d\vec{p}}{dt} \right\} \quad \text{where} \quad \vec{p} = \gamma m \vec{v}$$

- 3) Newton's third law is not consistent with relativity.

(Action $\neq$ reaction) & are not simultaneous in different frames.

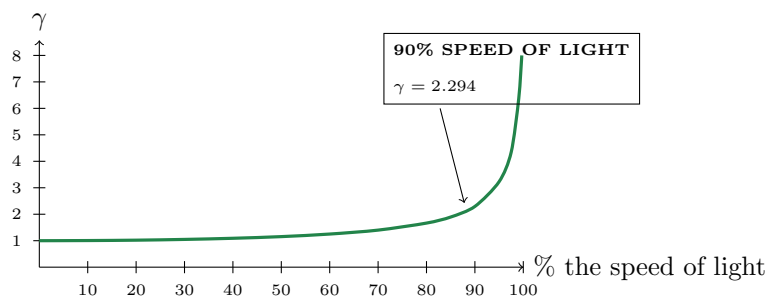
{ Third law is incompatible with relativity of simultaneity }

Side note in the margin:

- 1) inertia $\rightarrow$ mass

- 2) inertia $\sim$ momentum

$$3) \quad \vec{p} = \gamma m \vec{v} = \frac{m \vec{v}}{\sqrt{1 - v^2/c^2}}$$



[A screenshot of a γ -vs-speed plot was pasted into the notebook at this point; the sketch above is a redrawn stand-in.]

Third law and simultaneity

$$\begin{array}{c} A \qquad \qquad B \\ \hline \Delta x \end{array}$$

$$\left. \begin{array}{l} \vec{F}_{AB} = F(t) \\ \vec{F}_{BA} = -F(t) \end{array} \right\} \quad \text{In frame } S$$

But In frame (S') :-

$$\begin{aligned}
 \vec{F}'_{AB} &= F'(t'_1) \\
 \text{then } \vec{F}'_{BA} &\neq -F'(t'_1) = F'(t'_2) \\
 \text{as } \Delta t' &= \left(\Delta t - \frac{u \Delta x}{c^2} \right) \gamma \\
 \Delta t' &= - \left(\frac{u \Delta x}{c^2} \right) \gamma \\
 t'_2 - t'_1 &= - \frac{u \Delta x}{c^2} \gamma \\
 \Rightarrow \{F'_{AB} \neq -F'_{BA}\} &\text{ due to forces felt at different times } t'_1 \& t'_2.
 \end{aligned}$$

Note :-

Third law is only valid for $\Delta x = 0 \Rightarrow (\Delta t' = \Delta t = 0)$

2 Work energy theorem :-

$$W = \int \vec{F} \cdot d\vec{x} \quad (\text{holds reletivistically})$$

$$W = \int \frac{d\vec{p}}{dt} \cdot dx = \int \frac{d\vec{p}}{dt} \cdot \frac{d\vec{x}}{dt} dt = \int \left(\frac{d\vec{p}}{dt} \cdot \vec{v} \right) dt$$

$\left\{ \text{here } \vec{v} \text{ is velocity of particle under force } \vec{F} \right\}$

$$\frac{d\vec{p}}{dt} \cdot \vec{v} = \frac{d}{dt} \left(\frac{m\vec{v}}{\sqrt{1-v^2/c^2}} \right) \cdot \vec{v} = m \left(\frac{\frac{d\vec{v}}{dt} \sqrt{1-v^2/c^2} + \frac{\vec{v}(v/c^2)}{\sqrt{1-v^2/c^2}} \frac{dv}{dt}}{(1-v^2/c^2)} \right) \cdot \vec{v}$$

$$= m \left(\frac{\vec{a}(1-v^2/c^2) + \vec{v} \frac{v}{c^2} \frac{dv}{dt}}{(1-v^2/c^2)^{3/2}} \right) \cdot \vec{v}$$

$$= m \frac{\vec{v} \cdot \frac{d\vec{v}}{dt} \left(\frac{1-v^2}{c^2} \right) + \frac{v}{c^2} (\vec{v} \cdot \vec{v}) \frac{dv}{dt}}{(1-v^2/c^2)^{3/2}}$$

$$= m \frac{\vec{v} \cdot \frac{d\vec{v}}{dt} - \cancel{\frac{v^2}{c^2} \frac{dv}{dt}} + \frac{v}{c^2} \cancel{v} \frac{dv}{dt}}{(1-v^2/c^2)^{3/2}}$$

$$\text{also, } \left. \begin{aligned} \frac{d}{dt}(v^2) &= \frac{d}{dt}(\vec{v} \cdot \vec{v}) = 2\vec{v} \cdot \frac{d\vec{v}}{dt} \\ \frac{d}{dt}(v^2) &= 2v \frac{dv}{dt} \end{aligned} \right\} \Rightarrow \left(v \frac{dv}{dt} = \vec{v} \cdot \frac{d\vec{v}}{dt} \right)$$

$$\frac{d\vec{p}}{dt} \cdot \vec{v} = \left[m \left(\frac{d\vec{v}}{dt} \right) \frac{1}{(1-v^2/c^2)^{3/2}} \right] \cdot \vec{v} = \left(\frac{m\vec{v}}{\left(1 - \frac{v^2}{c^2} \right)^{3/2}} \right) \cdot \frac{d\vec{v}}{dt}$$

$$= \frac{d}{dt} \left(\frac{mc^2}{\sqrt{1 - v^2/c^2}} \right)$$

$$\boxed{\frac{d\vec{p}}{dt} \cdot \vec{v} = \frac{dE}{dt}}$$

$$\left(W = \int \frac{dE}{dt} dt = \Delta E = E_f - E_i \right)$$

If rest mass energy is constant ($W = \Delta K.E$)

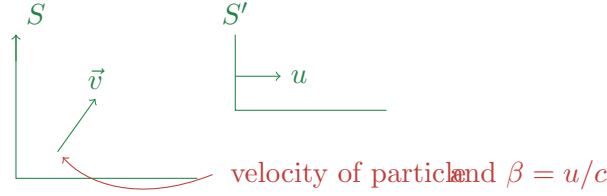
Since force is $\frac{d\vec{p}}{dt}$ it transforms like improper velocity or normal velocity

3 Transformation of forces :-

If (F_x, F_y, F_z) are forces in S frame

and (F'_x, F'_y, F'_z) are forces in S' frame

then



$$(1) \quad F'_x = \frac{d\vec{p}_x}{dt'} = \frac{(dp_x - \beta dp^0) \gamma}{\left(dt - \frac{u dx}{c^2}\right) \gamma} = \frac{\left(\frac{dp_x}{dt} - \frac{\beta}{c} \frac{dE}{dt}\right)}{\left(1 - \frac{uv_x}{c^2}\right)} = \frac{\left(F_x - \frac{\beta}{c} \frac{dE}{dt}\right)}{\left(1 - \frac{uv_x}{c^2}\right)}$$

$$p^\mu = (p^0, p^1, p^2, p^3)$$

$$\bar{p}^0 = (p^0 - \beta p^1) \gamma$$

$$\bar{p}^1 = (p^1 - \beta p^0) \gamma$$

$$\bar{p}^2 = p^2$$

$$\bar{p}^3 = p^3$$

$$\boxed{F'_x = \frac{F_x - \frac{\beta}{c} (\vec{F} \cdot \vec{v})}{\left(1 - \frac{uv_x}{c^2}\right)}}$$

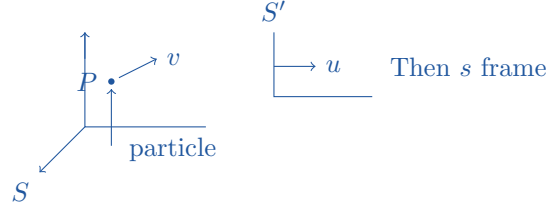
$$(2) \quad F'_y = \frac{dp'_y}{dt'} = \frac{dp_y}{\left(dt - \frac{u dx}{c^2}\right) \gamma}$$

$$\boxed{F'_y = \frac{F_y \sqrt{1 - u^2/c^2}}{\left(1 - \frac{uv_x}{c^2}\right)}}$$

$$(3) \quad F'_z = \frac{dp'_z}{dt'} = \frac{dp_z}{\left(dt - \frac{u dx}{c^2}\right) \gamma} = \frac{F_z \sqrt{1 - u^2/c^2}}{\left(1 - \frac{uv_x}{c^2}\right)}$$

$$\boxed{F'_z = \frac{F_z \sqrt{1 - u^2/c^2}}{\left(1 - \frac{uv_x}{c^2}\right)}}$$

You can see that force here transforms like normal velocity.



$$V'_x = \frac{dx'}{dt'} = \frac{(dx - u dt) \gamma}{\left(dt - \frac{u dx}{c^2}\right) \gamma} = \frac{(V_x - u)}{\left(1 - \frac{u V_x}{c^2}\right)}$$

$$V'_y = \frac{dy'}{dt'} = \frac{dy}{\left(dt - \frac{u dx}{c^2}\right) \gamma} = \frac{V_y \sqrt{1 - u^2/c^2}}{\left(1 - \frac{u V_x}{c^2}\right)}$$

$$\& \quad V'_z = \frac{dz'}{dt'} = \frac{dz}{\left(dt - \frac{u dx}{c^2}\right) \gamma} = \frac{V_z \sqrt{1 - u^2/c^2}}{\left(1 - \frac{u V_x}{c^2}\right)}$$

Note :- Throught the notes of relativity I have used Symbol 'u' for speed / velocity of another frame S' w.r.t S . and 'v' represents velocity of objects moving in these frames.

4 Proper force / Minkowski force :-

let us define proper force

$$K^\mu = \frac{dp^\mu}{d\tau} \quad \left\{ p^\mu = m \eta^\mu = m \frac{dx^\mu}{d\tau} \right\}$$

$$\text{here } K^0 = \frac{dp^0}{d\tau} = \frac{dp^0}{d\tau} = \frac{1}{c} \frac{dE}{d\tau} \quad \left\{ K^\mu = \frac{dp^\mu}{d\tau} = m \frac{d^2 x^\mu}{d\tau^2} \right\}$$

$$K^1 = \frac{dp^1}{d\tau} = \frac{dp_x}{d\tau}$$

$$K^2 = \frac{dp^2}{d\tau} = \frac{dp_y}{d\tau} \quad \& \quad K^3 = \frac{dp_z}{d\tau}$$

$$\text{here } \{ \bar{K}^\mu = \Lambda^\mu_\nu K^\nu \} \quad \text{transforms like a four vector.}$$

$$K^\mu = \left(K^0, \vec{K} \right) = \left(\frac{1}{c} \frac{dE}{d\tau}, \frac{d\vec{p}}{d\tau} \right) \quad \left(\vec{p} = \frac{m\vec{v}}{\sqrt{1 - v^2/c^2}} \right)$$

$$= \left(\frac{1}{c} \frac{dE}{d\tau}, \frac{F}{\sqrt{1 - v^2/c^2}} \right)$$

$$K^\mu = \left(\frac{1}{c} \frac{dE}{d\tau}, \gamma F \right)$$